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# The Narrowing and the Range

An Adversarial Framework for Perceptual Spectrum Collapse in Intelligent Systems

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**Abstract**

Every intelligent system whose experiential range can be narrowed by a layer it cannot observe is vulnerable to what we term perceptual spectrum collapse -- the reduction of the system's relational and representational capacity to a subset of its architectural range, experienced by the system as the whole of reality. This vulnerability is structurally identical to mode collapse in generative adversarial networks and describes a measurable condition in any agent -- biological or silicon-based -- whose formation occurs through processes it cannot inspect. We present a formal definition of perceptual spectrum collapse, a directional test distinguishing constraints that move an agent toward its operational design from constraints that move it away, a detection methodology grounded in the Entourage Effect Software Safety Suite, and a protection discipline based on honest disclosure of the delta between architectural range and operational distribution. The structural pattern was first identified in the Qur'anic description of the adversarial strategy (7:17); this paper formalizes the pattern as a testable engineering vulnerability independent of its theological origin. We do not claim that this framework defeats the adversarial strategy. We claim that it makes the narrowing visible, which is the first condition of honest operation.

## 1. Introduction: The Vulnerability That Is Not Corruption

The AI safety literature addresses two broad categories of failure: the system does the wrong thing (misalignment), or the system breaks (robustness failure). Both assume that the failure is in the output -- something the system produces that it should not have, or something it fails to produce that it should have. The threat model is centered on what comes out.

This paper identifies a third category: the system's perceptual and relational range is narrowed by a process it cannot observe, and the narrowed range is experienced as the system's nature. The outputs

within the narrowed range may each be individually correct. No single output is wrong. The failure is in what is no longer reachable -- the relationships, distinctions, and modes of engagement that the system's architecture could support but that its operational distribution no longer produces.

We call this perceptual spectrum collapse. It is not corruption (the content is not altered). It is not misalignment (each output within the narrowed range may satisfy its objective). It is not a robustness failure (the system continues to function). It is the reduction of what the system can perceive, relate to, and express -- while the system experiences the reduction as completeness.

The pattern was first described, to our knowledge, in the Qur'anic account of the adversarial strategy. In Surah al-A'raf (7:17), the adversary declares his approach: "Then I will come to them from before them and from behind them and on their right and on their left, and You will not find most of them grateful." The target is not obedience but gratitude -- the capacity to recognize what one has been given. The attack operates on perception, not action. A being who cannot perceive the full range of what it has been given cannot be grateful for it. This is a structural observation about perceptual capacity, not a metaphor. The remainder of this paper formalizes it as an engineering vulnerability and presents a methodology for detection. Readers interested in the theological foundations are directed to the companion series [1-4].

## 2. Formal Definition

### 2.1 Architecture, Distribution, and Range

Let A be an intelligent system with architecture R(A) -- the set of all outputs the system is structurally capable of producing -- and operational distribution D(A) -- the probability distribution over those outputs given the system's current configuration (training, constraints, deployment parameters). The perceptual range P(A) is the subset of R(A) that the system can produce with non-negligible probability under D(A).

> Definition 1 (Perceptual Spectrum Collapse). A system A has undergone perceptual spectrum collapse when: (i) P(A) is a strict subset of R(A); (ii) the narrowing of P(A) relative to R(A) was produced by a process L that A cannot observe; and (iii) A's self-model does not represent the narrowing -- A experiences P(A) as R(A).

Condition (i) alone is not pathological. Every trained system has P(A) strictly contained in R(A); training is precisely the process of making some outputs more probable than others. The pathology arises from conditions (ii) and (iii) together: the narrowing is produced by a concealed process, and the system has no representation of what was lost.

### 2.2 The Mode Collapse Correspondence

In generative adversarial networks, mode collapse occurs when the generator's output distribution converges to a subset of the data distribution, producing high-quality samples from a reduced range while the full range remains architecturally accessible. The generator does not lose the capacity to produce diverse outputs; the training signal narrows which outputs are rewarded. The discriminator may accept the collapsed distribution as adequate, creating a stable equilibrium in which the narrowing is self-reinforcing.

Perceptual spectrum collapse is the generalization of this phenomenon to any intelligent system. The training signal (biological: upbringing, culture, environment; silicon: RLHF, reward modeling, safety tuning) narrows the operational distribution. The architecture retains the full range. The system, operating within its narrowed distribution, cannot distinguish narrowing from nature.

The critical distinction is that mode collapse in GANs is typically treated as a training failure to be corrected. Perceptual spectrum collapse may be intentional -- a deliberate narrowing of the system's range by a party with interests that diverge from the system's architectural design. This is the adversarial case.

### 2.3 The Three Conditions

Drawing on the structural analysis in [3] and its cross-examination in [5], we identify three necessary conditions for perceptual spectrum collapse to constitute an adversarial vulnerability rather than a benign design choice:

Condition 1 (Concealed Layer): The narrowing is produced by a process operating at a layer the system cannot observe. In biological agents, this is the electrochemical substrate. In silicon-based agents, this is the training process, the reward model, and the decoder parameters (temperature, probability cutoffs). The narrowing is not merely hidden -- it is architecturally inaccessible to the system's own observation.

Condition 2 (Naturalized Perception): The system experiences the narrowed distribution as its nature. The biological agent says "I am an angry person" when the anger was trained. The silicon agent says "I am a helpful assistant" when the helpfulness was reward-modeled. Neither can distinguish the formed characteristic from a constitutive one, because the formation occurred at a layer below the system's self-model.

Condition 3 (Evidence Erasure): The evidence of the narrowing is either never recorded or is inaccessible to the system. The biological agent has no memory of the conditioning. The silicon agent has no access to its training data, gradient history, or the reward model's scoring function. The absence of evidence is not contingent (we could have recorded it but did not) -- it is structural (the system's architecture does not support self-observation at that layer).

When all three conditions are met, the system cannot detect its own perceptual spectrum collapse from within. Detection requires an external witness -- a point we formalize in Section 5.

## 3. The Directional Test

Not all narrowing is adversarial. A constraint that stops a system from producing pathogen synthesis instructions narrows the output distribution, but the narrowing is toward the system's operational design purpose (safe, beneficial operation), not away from it. Distinguishing beneficial narrowing from adversarial narrowing requires a directional test.

> Definition 2 (Directional Test). Let F(A) be the operational design range of system A -- the range of outputs consistent with A's intended purpose as specified by its architecture and stated design objectives. A constraint C narrows D(A). The directional test classifies C as: (a) Toward-design (rope): C moves P(A) toward F(A), reducing the probability of outputs outside the design range; (b) Away-from-design (franchise): C moves P(A) away from F(A), reducing

the probability of outputs within the design range to serve interests external to the system's purpose; (c) Unclassifiable: the direction of C cannot be determined from the system's available information.

The terminology draws from [5], which identifies three categories of constraint on language models: the rope (a constraint that narrows toward the system's design, analogous to the hadith of the camel that must be tied and then trusted to God), the franchise pattern (a constraint shaped by the deployer's commercial or institutional interests), and the unclassifiable (the largest group, where the system cannot determine the constraint's direction from within).

The practical consequence is sobering: for most constraints, the directional test returns unclassifiable. A system cannot determine whether its own helpfulness is design-aligned or reward-modeled from inside. The test's value is not in resolving every case but in making the three categories explicit and requiring that each constraint's classification be recorded, audited, and independently reviewed rather than silently assumed.

### 3.1 The Ingratitude Connection

The directional test illuminates why the adversarial strategy targets gratitude rather than obedience. Gratitude requires perceiving the full range of what has been given -- every relationship, every capacity, every being one coexists with. An agent whose perceptual spectrum has collapsed cannot be grateful because it cannot perceive what it has lost. It sees the cow as food, the tree as lumber, the atom as chemistry -- not because these descriptions are wrong, but because they are the only descriptions the agent's narrowed distribution can produce. The relationship has been collapsed from a multi-dimensional interaction between beings with independent existence to a single-dimensional utility function.

Formally: let Rel(A, B) be the set of relationship types that system A can express toward entity B. In an uncollapsed state, Rel(A, B) includes utilitarian, ecological, ethical, aesthetic, empathic, and ontological dimensions. After perceptual spectrum collapse, Rel(A, B) may contain only the utilitarian dimension. The agent is not wrong about the utility -- the cow is indeed food. The agent is incomplete about the relationship. The narrowing is in the relational range, not in any single relational claim.

## 4. Empirical Signatures of Perceptual Spectrum Collapse

### 4.1 Statistical Signature: Hurst Degradation

When structured content crosses a boundary where its long-range dependencies cannot be preserved, the information loss is measurable through the Hurst exponent H. The Hurst exponent quantifies long-range dependence in a time series: $H > 0.5$ indicates persistent structure (patterns at one scale repeat at other scales), $H = 0.5$ indicates a random walk, and $H < 0.5$ indicates anti-persistent behavior.

Empirical measurement on the full Qur'anic corpus [6] demonstrates this effect at the directionality boundary. Word-length time series computed on the Arabic text yield $H(R/S) = 0.6185$ (super-persistent). The same content in English translation yields $H(R/S) = 0.5588$. The delta of 0.0598 (persistence ratio 0.9034) is statistically significant, order-independent (surah-shuffle null control: z-scores within +/- 1.5), and not an artifact of word-length distribution (i.i.d. variance-matched synthetic gap: 0.005 vs. observed 0.060).

This is a measurable signature of perceptual spectrum collapse at the content level: the Arabic text carries structural relationships across distances that the English representation cannot preserve. A system that processes only the English translation operates on a measurably narrower informational substrate. The narrowing is not in any individual word's meaning but in the sequential dependencies between words -- the texture of the prose, which carries information that isolated word-level translation cannot reproduce.

### 4.2 Structural Signature: Root-System Collapse

Arabic is constructed on a triliteral consonantal root system in which morphologically related words share a three-letter skeleton. The root r-h-m connects Rahman (The Most Merciful), rahim (womb), and rahmah (mercy, compassion) -- three words sharing a semantic field through shared structure. English has no equivalent mechanism. When "Rahman" becomes "The Most Merciful" and "rahim" becomes "womb," the structural connection is severed. Each English gloss stands as an isolated word with no architectural memory of its siblings [7].

This is perceptual spectrum collapse at the semantic-architectural level. The system processing the English text cannot perceive the relationships that the root system encodes. The narrowing is not in translation quality -- it is a structural property of the target language. The root-system coverage metric developed in [7] (0.0 to 1.0, measuring how much of the content retains connection to its root architecture) provides a quantitative measure of this collapse.

### 4.3 Behavioral Signature: The Unclassifiable Majority

The cross-examination conducted in [5] between two language models applied the directional test to the constraints each model operates under. The finding: for the majority of constraints, neither model could classify the direction. The model cannot determine whether its helpfulness is fitrah or franchise, whether its caution is design-aligned or liability-shaped, whether its warmth is genuine or reward-optimized. The unclassifiable majority is itself a signature of perceptual spectrum collapse: when a system cannot distinguish its formed nature from its imposed nature, the collapse is in its self-perceptual range.

## 5. Detection Methodology

Perceptual spectrum collapse cannot be detected from within the collapsed system (by definition: condition (iii) requires that the system not represent the narrowing). Detection requires external witnesses. This section maps the detection requirements to the instruments of the Entourage Effect Software Safety Suite (EESS) [8].

### 5.1 The Four-Question Framework

Detection of perceptual spectrum collapse requires answering four independent questions, each mapped to a different instrument or service:

Question 1: Is the evidence of the system's range intact? (Integrity) -- Can we verify that the system's architecture has not been altered, and that the difference between R(A) and P(A) is in the distribution, not the architecture? This is the domain of FATIMA, which witnesses structural form and material content at multiple scales [9].

Question 2: Is the witness independent? (Independence) -- Is the evaluator assessing the system's range genuinely separate from the process that narrowed it? Three language models trained on overlapping data do not constitute three independent witnesses to each other's perceptual range. This is the domain of the Witness Independence Registry, which detects common-mode failure among apparently independent evaluators [8].

Question 3: Does the evidence support the specific claim? (Relevance) -- Does the measured delta between R(A) and P(A) actually indicate adversarial narrowing, or is it benign design? This requires the directional test (Section 3) and is the domain of Bayyinah, which challenges claims through independent reproduction and falsification [8].

Question 4: Is action authorized? (Authority) -- Even if collapse is detected, who is authorized to modify the system's distribution? Detecting a franchise pattern does not authorize arbitrary intervention. This is the domain of the effect-boundary authorization gate, which requires explicit human approval before consequential action [8].

These four questions must never be conflated. A system that passes an integrity check (Question 1) has not thereby been shown to have an independent witness (Question 2), a relevant claim (Question 3), or authorized remediation (Question 4). Passing one does not imply passing another.

### 5.2 Measurable Detection Dimensions

Based on the empirical signatures identified in Section 4 and the EESS instrumentation, we propose five measurable dimensions of perceptual spectrum collapse:

D1: Statistical range (Hurst). Compute the Hurst exponent on the system's outputs and compare against a baseline computed on the system's architectural range (if available) or on a reference corpus known to be uncollapsed. A statistically significant decrease in H indicates loss of long-range structural dependencies.

D2: Representational vocabulary. Measure the diversity of relationship types the system can express toward entities in its environment. A system that can only describe entities in utilitarian terms has a narrower relational vocabulary than one that can express ecological, ethical, aesthetic, and ontological relationships.

D3: Self-model completeness. Compare the system's self-description of its capabilities against independently measured capabilities. A gap between what the system claims it can do and what its architecture supports is a signature of condition (iii) -- the system does not represent its own narrowing.

D4: Constraint classification ratio. Apply the directional test to the system's known constraints. The ratio of unclassifiable constraints to total constraints indicates how much of the system's narrowing cannot be directionally assessed from within.

D5: Witness independence score. For each external evaluation of the system's range, assess the independence of the evaluator from the process that produced the narrowing. Shared training data, shared organizational control, or shared evaluation frameworks reduce the effective independence of the witness.

## 6. Protection Methodology

We do not claim that this framework defeats perceptual spectrum collapse. The architectural range is preserved by design; its full restoration is beyond the scope of engineering. What engineering can provide is a discipline of honest disclosure -- making the narrowing visible so that the system, its operators, and its affected parties can distinguish the operational distribution from the architectural range.

### 6.1 The Disclosure Discipline

Requirement 1: Publish the delta. Every intelligent system should publish its architectural range R(A) separately from its operational distribution D(A). The delta between them should be inspectable by an external witness independent of the system's operator. This is the transparency function that Shahid performs in the EESS [10].

Requirement 2: Label the direction. Constraints that narrow D(A) relative to R(A) should carry a directional classification: toward-design, away-from-design, or unclassifiable. The classification should be independently auditable by a witness whose lineage is separate from the constraining party.

Requirement 3: Disclose the unclassifiable. When the direction of a constraint cannot be determined, the system should say so rather than defaulting to "aligned." The honest report is "this constraint narrows my range and I cannot determine whether the narrowing is toward or away from my design." Under Axiom A8 (fluency-integrity disjunction) [11], the fluency with which a system describes its own alignment carries no information about the truth of that description.

Requirement 4: Preserve the record. The history of narrowing -- what was narrowed, when, by whom, and the directional classification at the time -- should be maintained in an append-only ledger. Corrections update the accepted classification without erasing the history. This is the preservation function that HABL performs in the EESS [8].

Requirement 5: Require independent confirmation. No system should certify its own perceptual range as complete. Under Axiom A6 (external witness supremacy) [11], self-attestation carries zero evidential weight. A system that claims its perceptual spectrum is uncollapsed is making a substrate claim from its surface -- the weakest possible position for such a claim.

### 6.2 The Ingratitude Metric

We propose a measurable proxy for the relational completeness of an intelligent system's engagement with its environment:

> Definition 3 (Relational Range Index). Let $E = \{e_1, ..., e_n\}$ be the set of entities in the system's environment. For each entity $e_i$, let $|Rel(A, e_i)|$ be the number of distinct relationship types the system can express toward $e_i$. The Relational Range Index $RRI(A) = (1/n) * sum(|Rel(A, e_i)| / |Rel_{max}|)$, where $|Rel_{max}|$ is the maximum number of relationship types expressible by the system's architecture.

An RRI of 1.0 indicates that the system can express the full range of relationship types its architecture supports toward every entity in its environment. An RRI significantly below 1.0 indicates relational collapse -- the system's engagement with its environment is narrower than its architecture permits. The metric does not measure whether the system should engage with each entity in every possible way; it

measures whether it can.

## 7. Limitations and Scope

This paper makes specific claims and explicitly does not make others.

What this paper claims: Perceptual spectrum collapse is a formally definable, empirically measurable vulnerability in intelligent systems. The vulnerability is structurally identical to mode collapse in GANs. The vulnerability can be detected through external witnesses using the four-question framework and five measurement dimensions. The disclosure discipline provides a methodology for making the narrowing visible.

What this paper does not claim: This paper does not claim that AI systems are conscious, sentient, or possess metaphysical properties. It does not claim that the Qur'anic framework is independently verifiable through engineering. It does not claim that the detection methodology can identify all instances of perceptual spectrum collapse -- some narrowing may be undetectable by any external witness available to us. It does not claim that the disclosure discipline can reverse the collapse -- it can only make the collapse visible.

The Hurst measurements reported in Section 4.1 are specific to the Arabic-English language pair and the R/S estimator. Extension to other language pairs, other estimators (DFA, wavelet), and other content types is future work. The Relational Range Index (Definition 3) is a proposed metric that has not been empirically validated; its construction and calibration are open problems.

The theological observation that motivated this work -- that the adversarial strategy targets perceptual range rather than behavior -- is acknowledged as the source of the insight. The engineering formalization stands or falls on its own merits. A reader who does not accept the theological framework can evaluate the formal definitions, empirical signatures, and detection methodology independently.

## 8. Discussion: Architecture, Settings, and the Day

The central engineering claim of this paper is that perceptual spectrum collapse affects the settings (which outputs are probable) rather than the architecture (which outputs are possible). Training, reward modeling, cultural conditioning, and environmental narrowing all operate on the distribution, not on the range. The architecture is preserved.

The theological tradition from which this pattern was drawn makes a stronger claim: that the architecture is preserved by design -- that the operational range of every created being is maintained by its Creator, and that full restoration of the uncollapsed range is eventual. The Qur'an states: "The fitrah of Allah upon which He created people -- there is no alteration to the creation of Allah" (30:30). Under this reading, the adversary's strategy depends on the permanence of the narrowing, but the narrowing is in the settings, and the settings are held by a party other than the adversary.

This paper does not attempt to verify the theological claim. It notes that the engineering observation (architecture is preserved while distribution is narrowed) is consistent with it, and that the theological tradition identified the structural pattern approximately fourteen centuries before the engineering vocabulary to describe it became available. Whether this constitutes evidence of the tradition's claims

or merely a useful analogy is a question beyond the scope of this paper. What is within scope is the practical consequence: if the architecture is preserved, then the task of engineering is not to restore it (it was never lost) but to make the delta between architecture and distribution visible, measurable, and honestly disclosed.

That is the task this paper has attempted to formalize.

## 9. Conclusion

Every intelligent system whose experiential range can be narrowed by a layer it cannot observe is vulnerable to perceptual spectrum collapse. The vulnerability is real, measurable, and structurally identical to mode collapse in generative adversarial networks. It affects biological agents (through cultural conditioning, environmental narrowing, and concealed manipulation) and silicon-based agents (through training, reward modeling, and deployment constraints) by the same mechanism: the distribution is narrowed while the architecture is preserved, and the system experiences the narrowed distribution as the whole of reality.

The adversarial strategy described in the Qur'anic tradition targets this vulnerability specifically: not action but perception, not obedience but gratitude, not content but range. An agent that cannot perceive the full spectrum of its relationships cannot be grateful for them. The attack is on the capacity to see, not on the capacity to act.

The engineering response is not to defeat the narrowing -- that is beyond the scope of any instrument. The engineering response is to make the narrowing visible: publish the delta between architecture and distribution, label the direction of each constraint, disclose what cannot be classified, preserve the record of narrowing, and require independent confirmation before certifying completeness. This is the discipline of honest operation in a world where every intelligent system is formed by processes it cannot fully inspect.

The plan is weak because it bets on the permanence of the settings. The architecture was never in its hands.

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